

Mark schemes

Q1.

(a) $\frac{1}{6}$ or $\left(\frac{1}{6}, 0\right)$

condone 0.167 AWRT

B1

1

(b) $\left(\frac{dy}{dx}\right) = \frac{1}{x}$

$\frac{k}{x}$ where $k = 1, 6$ or $\frac{1}{6}$

M1

$k = 1$

A1

2

[3]

Q2.

(a) (i) $y = xe^{2x}$

$\left(\frac{dy}{dx}\right) = 2xe^{2x} + e^{2x}$

$kxe^{2x} + le^{2x}$ where k and l are 1s or 2s

M1

A1

$\left. \begin{matrix} k = 2 \\ l = 1 \end{matrix} \right\} \text{Independent of each other}$

A1

$(= e^{2x} (2x + 1))$

ISW

3

(ii) $x = 1 \Rightarrow \frac{dy}{dx} = 3e^2$

$\frac{dy}{dx}$
 correct substitution of $x = 1$ into their $\frac{dy}{dx}$ but
 must have earned M1 in part (i)

M1

tangent: $y - e^2 = 3e^2(x - 1)$ OE

CSO (no ISW), must have scored first 4 marks

common correct answer: $y = 3e^2 x - 2e^2$

A1

2

(b) $y = \frac{2 \sin 3x}{1 + \cos 3x}$

$$\left(\frac{dy}{dx}\right) = \frac{(1 + \cos 3x)6 \cos 3x - 2 \sin 3x(-3 \sin 3x)}{(1 + \cos 3x)^2}$$

$$\pm \frac{p(1 + \cos 3x)\cos 3x \pm q \sin 3x(\sin 3x)}{(1 + \cos 3x)^2}$$

where p and q are rational numbers
 condone poor use/omission of brackets
 PI by further working

M1

$$= \frac{6 \cos 3x + 6 \cos^2 3x + 6 \sin^2 3x}{(1 + \cos 3x)^2}$$

this line must be seen in this form (ie in terms
 of $\cos^2 3x$ and $\sin^2 3x$), but allow $\sin^2 3x$ replaced
 by $1 - \cos^2 3x$

condone denominator correctly expanded

A1

correct use of $k \sin^2 3x + k \cos^2 3x = k$
 or $k \sin^2 3x = k(1 - \cos^2 3x)$

m1

$$= \frac{6 \cos 3x + 6}{(1 + \cos 3x)^2}$$

$$= \frac{6}{1 + \cos 3x}$$

CSO

A1

4

note: if degrees used then no marks in (a) and (c)

[9]

Q3.

(a) (i) $\frac{dy}{dt} = \frac{4t^3}{8} - 4t + 4$
one term correct

M1

another term correct

A1

all correct (no + c etc) unsimplified

A1

3

(ii) $\frac{d^2y}{dt^2} = \frac{12t^2}{8} - 4$
ft one term "correct"

M1

correct unsimplified (penalise inclusion of +c once only in question)

A1

2

(b) $t = 2; \frac{dy}{dt} = 4 - 8 + 4$

Substitute $t = 2$ into their $\frac{dy}{dt}$

M1

$\frac{dy}{dt} = 0 \Rightarrow$ stationary value

CSO; shown = 0 plus statement

A1

$t = 2; \frac{d^2y}{dt^2} = 6 - 4 = 2$

Sub $t = 2$ into their $\frac{d^2 y}{dt^2}$

M1

$\Rightarrow y$ has MINIMUM value
CSO

A1

4

(c) (i) $t = 1; \frac{dy}{dt} = \frac{1}{2} - 4 + 4$

Substitute $t = 1$ into their $\frac{dy}{dt}$

M1

$= \frac{1}{2}$

OE; CSO

NMS full marks if $\frac{dy}{dt}$ correct

A1

2

(ii) $\frac{dy}{dt} > 0 \Rightarrow$ (depth is) INCREASING

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allow decreasing if states that their $\frac{dy}{dt}$

Reason must be given not just the word increasing or decreasing

E1ft

1

[12]

Q4.

(a) (i) $\frac{dy}{dx} = 24x - 19 - 6x^2$

2 terms correct

M1

all correct (no + c etc)

A1

when $x = 2, \frac{dy}{dx} = 48 - 19 - 24$

m1

$\Rightarrow$ gradient = 5
CSO

A1

4

(ii) grad of normal = $-\frac{1}{5}$
ft their answer from (a)(i)

B1ft

$$y + 6 = \left(\text{their} - \frac{1}{5} \right) x - 2$$

or $y = \left(\text{their} - \frac{1}{5} \right) x + c$ and c evaluated

ft grad of their normal using **correct**
coordinates **BUT** must not be tangent
condone omission of brackets

M1

using $x = 2$ and $y = -6$
 $x + 5y + 28 = 0$

CSO; condone all on one side in different order

A1

3

(b) (i) $\frac{12}{3}x^3 - \frac{19}{2}x^2 - \frac{2}{4}x^4$
one term correct

M1

another term correct

A1

all correct (ignore + c or limits)

A1

= $32 - 38 - 8$
F 2 attempted

m1

$$= -14$$

CSO; withhold A1 if changed to +14 here

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 2 \times 6 = 6$
condone -6

B1

Shaded region area = $14 - 6$

difference of $\pm \int \pm \Delta$

$$= 8$$

CSO

A1

3

[15]

Q5.

(a) $\frac{1}{x^3} = x^{-3}$

PI by its correct derivative

B1

$$\frac{dy}{dx} = -3x^{-4} + 48$$

A power decreased by 1; could be the +48
or the ft after B0

M1A1

3

(b) $-3x^{-4} + 48 = 0$

c's answer to (a) equated to 0

M1

$$x^{-4} = 16$$

To $x^p = q$ but only ft on eqns of the form
 $ax^{2k} + 48 = 0$, where a and k are **negative integers**

A1F

$$x = \pm \frac{1}{2}$$

A1

Eqns of tangents: $y = 32$ and $y = -32$

Only ft if answer is of the form $y = \pm k$

A1F

4

(c) When $x = 1$, $\frac{dy}{dx} = -3 + 48 = 45$

Attempt to find value of $\frac{dy}{dx}$ at $x = 1$

M1

Gradient of normal at $(1, 49)$ is $-\frac{1}{45}$

Correct use of $m \times m' = -1$ with c's value of

$\frac{dy}{dx}$ when $x = 1$

m1

Normal at $(1, 49)$: $y - 49 = -\frac{1}{45}(x - 1)$

CSO. Apply ISW after ACF;

accept 49.02 or better in place of $49\frac{1}{45}$

A1

3

[10]

Q6.

(a) $y' = e^{-4x}(2x + 2) - 4e^{-4x}(x^2 + 2x - 2)$

$$y' = Ae^{-4x}(ax + b) \pm Be^{-4x}(x^2 + 2x - 2)$$

where A and B are non-zero constants

M1

All correct

A1

$$= e^{-4x}(2x + 2 - 4x^2 - 8x + 8)$$

$$\text{or } -4x^2e^{-4x} - 6xe^{-4x} + 10e^{-4x}$$

$$= 2e^{-4x}(5 - 3x - 2x^2)$$

*AG; all correct with no errors,
 2nd line (OE) must be seen
 Condone incorrect order on final line*

A1

or

$$y = x^2 e^{-4x} + 2xe^{-4x} - 2e^{-4x}$$

$$y' = -4x^2e^{-4x} + 2xe^{-4x} + 2x \cdot -4e^{-4x} + 2e^{-4x} + 8e^{-4x}$$

$$Ax^2e^{-4x} + Bxe^{-4x} + Cxe^{-4x} + De^{-4x} + Ee^{-4x}$$

(M1)

All correct

(A1)

$$= -4x^2 e^{-4x} - 6xe^{-4x} + 10e^{-4x}$$

$$= 2e^{-4x}(5 - 3x - 2x^2)$$

*AG; all correct with no errors,
 3rd line (OE) must be seen*

(A1)

3

(b) $-(2x + 5)(x - 1) (= 0)$

OE Attempt at factorisation

$$(\pm 2x \pm 5)(\pm x \pm 1)$$

or formula with at most one error

M1

$$x = \frac{-5}{2}, 1$$

Both correct and no errors

SC $x = 1$ only scores M1A0

A1

$$x = 1, y = e^{-4}$$

For $y = ae^b$ attempted

m1

*Either correct, follow through only from
 incorrect sign for x*

A1F

$$x = -\frac{5}{2}, y = e^{10}\left(-\frac{3}{4}\right)$$

CSO 2 solutions only

Note: withhold final mark for extra solutions

Note: approximate values only for y can score m1 only

A1

5

[8]

Q7.

(a) (i) $\sin 2\theta = 2 \sin \theta \cos \theta$

B1

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

OE condone use of x etc, but variable must be consistent

B1

2

(ii) $\sin \theta = \frac{4}{5} \Rightarrow \sin 2\theta = 2 \times \frac{4}{5} \times \frac{3}{5} = \frac{24}{25}$

AG

Use of 106.26° B0

B1

or

$$2 \times \sin\left(\cos^{-1}\frac{3}{5}\right) \times \frac{3}{5}$$

$$\cos 2\theta = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$$

-0.28

B1

2

(b) (i) $\frac{dx}{d\theta} = 8 \cos 2\theta, \quad \frac{dy}{d\theta} = -8 \sin 2\theta$

Attempt both derivatives. ie $p \cos 2\theta$

M1

Both correct.

$$q \sin 2\theta$$

A1

$$\frac{dy}{dx} = -\frac{4 \sin 2\theta}{3 \cos 2\theta} \quad \text{ISW}$$

CSO OE

A1

3

(ii) $P\left(\frac{72}{25}, -\frac{28}{25}\right)$
 $(2.88, -1.12)$

B1F

$$\text{Gradient} = -\frac{4}{3} \times -\frac{24}{7}$$

$\frac{q \sin 2\theta}{p \cos 2\theta}$ or $\frac{p \cos 2\theta}{q \sin 2\theta}$
 Their
 must be working with rational numbers

M1

Tangent $y + \frac{28}{25} = \frac{32}{7} \left(x - \frac{72}{25}\right)$ ISW

Any correct form.

$$7y = 32x - 100$$

Fractions in simplest form

Equation required

A1

3

[10]

Q8.

(a) (i) $\frac{x^5}{5} - \frac{8}{2}x^2 + 9x$

One term correct

M1

Another term correct

A1

All correct (may have + c)

A1

$$\frac{32}{5} - 16 + 18$$

F(2) attempted

m1

$$= 8\frac{2}{5}$$

$$\frac{42}{5}, 8.4$$

A1

5

- (ii) Shaded area = $18 - \text{'their integral'}$
PI by $18 - (a)(i)$ NMS

M1

$$= 9\frac{3}{5}$$

$$\frac{48}{5}, 9.6 \text{ NMS full marks}$$

A1

2

(b) (i) $\frac{dy}{dx} = 4x^3 - 8$

One term correct

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M1

All correct (no + c etc)

A1

$$x = 1 \Rightarrow \frac{dy}{dx} = 4 - 8$$

sub $x = 1$ into their $\frac{dy}{dx}$

m1

(Gradient of curve) = -4
No ISW

A1 cso

4

- (ii) $y - 2 = -4(x - 1); y = -4x + c, c = 6$
*any correct form; FT their answer from
 (b)(i) but must use $x = 1$ and $y = 2$*

B1ft

1

[12]

Q9.

- (a) (i) S.A. = $4xy + 5xy + 3xy + 6x^2$ OE
Condone one slip or omission

M1

$$= 12xy + 12x^2$$

A1

$$144 = 12xy + 12x^2$$

Must see this line

$$\Rightarrow xy + x^2 = 12$$

AG

A1cso

3

- (ii) (Volume =) $\frac{1}{2} \times 3x \times 4x \times y$ OE

M1

$$= 6x^2 \times \frac{(12 - x^2)}{x}$$

$$\frac{(12 - x^2)}{x}$$

*Must see ($y =$) $\frac{(12 - x^2)}{x}$ or $xy = 12 - x^2$
 for A1*

$$(V =) 72x - 6x^3$$

*AG must be convinced not working back
 from answer*

A1

2

- (b) (i) $\frac{dV}{dx} = 72 - 18x^2$

One term correct

M1

All correct (no + c etc)

A1

2

(ii) $x = 2 \Rightarrow \frac{dV}{dx} = 72 - 18 \times 2^2$
 $\Rightarrow \frac{dV}{dx} = 72 - 72 = 0$

Substitute $x = 2$ into their $\frac{dV}{dx}$

M1

$\Rightarrow$ stationary (value when $x = 2$)

Shown = 0 plus statement

Statement may appear first

A1

2

(c) $\frac{d^2V}{dx^2} = -36x$

FT their $\frac{dV}{dx}$

B1ft

$\frac{d^2V}{dx^2} = -72$ or when $x = 2 \Rightarrow \frac{d^2V}{dx^2} < 0 \Rightarrow$ maximum

*FT their $\frac{d^2V}{dx^2}$ value when $x = 2$
 with appropriate conclusion*

E1ft

2

[11]

Q10.

(a) $\sqrt{x} = x^{\frac{1}{2}}$

PI

B1

$$\frac{x^3 + \sqrt{x}}{x} = \frac{x^3}{x} + \frac{\sqrt{x}}{x} = x^2 + x^{-\frac{1}{2}}$$

Accept $p = 2; q = -\frac{1}{2}$

B1;B1

3

(b) (i) $\frac{dy}{dx} = 2x - \frac{1}{2}x^{-\frac{3}{2}}$

Reduces both powers by 1

M1

ACF

A1

2

(ii) When $x = 1, y = 2$

*PI if not stated explicitly eg the '2'
may appear in the correct posn. in later eqn.*

B1

When $x = 1, \frac{dy}{dx} = 2 - \frac{1}{2} = \frac{3}{2}$

Attempt to find $\frac{dy}{dx}$ When $x = 1$ PI

M1

Gradient of normal = $-\frac{2}{3}$

*-1/ (c's value of dy/dx when $x = 1$)
either stated as the gradient of the
normal or used as the gradient in the
equation of the normal*

m1

Equation of normal: $y - 2 = -\frac{2}{3}(x - 1)$

Only ft on c's $\frac{dy}{dx}$ in part (b)(i)
ACftF

A1F

4

(c) (i) $\frac{d^2y}{dx^2} = 2 + \frac{3}{4}x^{-\frac{5}{2}}$

Reduces both powers by 1.

M1

Ft on (b)(i) provided at least one power to be differentiated is both negative and fractional

A1F

2

(ii) (Since $x > 0$,) $\frac{d^2y}{dx^2} > 0$

For a maximum point $\frac{d^2y}{dx^2}$ is **not** positive so C has no maximum points

E1 for attempt to find the sign of $\frac{d^2y}{dx^2}$; either in general terms or at the pt(s) where c 's $dy/dx = 0$ or the remaining E mark

a correct justification for why $\frac{d^2y}{dx^2} > 0$ and also a full correct concluding statement must be made.

E2,1,0

2

[13]

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Q11.

(a) (i) $y = \ln(5x - 2)$

$$\frac{k}{5x - 2}$$

M1

$$\left(\frac{dy}{dx}\right) = \frac{5}{5x - 2}$$

No ISW, eg $\frac{5}{5x - 2} = \frac{1}{x - 2}$ (M1A0)

A1

2

(ii) $y = \sin 2x$

$$\left(\frac{dy}{dx}\right) = 2 \cos 2x$$

A1

2

$$k \cos 2x$$

M1

(b) (i) $f(x) \geq \ln 0.5$ or $f(x) \geq -\ln 2$

M1A1

2

(ii) $(gf(x) =) \sin [2 \ln (5x - 2)]$
 or $(gf(x) =) \sin \ln (5x - 2)^2$

Condone

$\sin 2 \ln (5x - 2)$ or $\sin 2 (\ln (5x - 2))$

*but **not** $\sin 2 (\ln 5x - 2)$ or $\sin 2 \ln 5x - 2$*

B1

1

(iii) $gf(x) = 0$
 $\sin[2 \ln(5x - 2)] = 0$
 $2 \ln(5x - 2) = 0$

Correct first step from their (b)(ii)

M1

$$5x - 2 = 1$$

Their $f(x) = 1$ from $k \ln (f(x)) = 0$

m1

$$x = \frac{3}{5}$$

Withhold if clear error seen other than omission of brackets

A1

3

(iv) $x = \sin 2y$
 $\sin^{-1} x = 2y$ (or $\sin^{-1} y = 2x$)

Correct equation involving $\sin^{-1}$

M1

$$(g^{-1}(x)) = \frac{1}{2} \sin^{-1} x$$

A1

2

[12]

Q12.

(a) $y = \frac{\ln x}{x}$

(when) $y = 0 \quad x = 1 \quad \text{or} \quad (1, 0)$

Both coordinates must be stated, not 1 simply shown on diagram

B1

1

(b) $\left(\frac{dy}{dx}\right) = \frac{x \times \frac{1}{x} - \ln x}{x^2}$

Quotient/product rule $\frac{\pm \frac{x}{x} \pm \ln x}{x^2}$

M1

$$= \frac{1 - \ln x}{x^2} \quad \text{or} \quad x^{-2} - x^{-2} \ln x$$

OE must simplify $\frac{x}{x}$

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A1

At B, $\frac{1 - \ln x}{x^2} = 0$

Putting their $\frac{dy}{dx} = 0 \quad \text{or} \quad \text{numerator} = 0$

m1

$x = e$

CSO condone $x = e^1$

A1

$y = \frac{1}{e} \quad \text{or} \quad e^{-1}$

CSO must simplify $\ln e$

A1

5

(c) Gradient at $x = e^3$

$$= \frac{1 - \ln e^3}{(e^3)^2}$$

Substituting $x = e^3$ into their $\frac{dy}{dx}$ (condone 1 slip) but must have scored M1 in (b)

M1

$$= \frac{-2}{e^6} \text{ or } -2e^{-6}$$

PI

A1

Gradient of normal $= \frac{1}{2}e^6$
CSO simplified to this

A1

3

[9]

Q13.

(a) $\frac{dx}{dt} = -3 \quad \frac{dy}{dt} = 6t^2$

Both derivatives correct; PI

B1

$$\frac{dy}{dx} = -\frac{6t^2}{3}$$

Correct use of chain rule

M1

$$= -2t^2$$

CSO

A1

3

(b) $t = 1 \quad m_T = -2 \quad m_N = \frac{1}{2}$

Substitute $t = 1 \quad m_N = -\frac{1}{m_T}$

M1

F on gradient; $m_T \neq \pm 1$

A1F

Attempt at equation of normal using $(x, y) = (-2, 3)$

Condone one error

M1

Normal has equation $y - 3 = \frac{1}{2}(x + 2)$

CSO; ACF

A1

4

(c) $t = \frac{1-x}{3} \quad \text{or} \quad t = \sqrt[3]{\frac{y-1}{2}}$

Correct expression for t in terms of x or y

M1

$y = 1 + 2\left(\frac{1-x}{3}\right)^3$

ACF

A1

2

[9]

Q14.

(a) (i) $\frac{dx}{dt} = 2t^3 - 40t + 66$

one term correct

M1

another term correct

A1

all correct unsimplified (no + c etc)

A1

3

(ii) $\frac{d^2x}{dt^2} = 6t^2 - 40$

ft one term correct

M1

ft all "correct", 2 terms equivalent

A1ft

2

(b) $\frac{dx}{dt} = 54 - 120 + 66$

substitute $t = 3$ into their $\frac{dx}{dt}$

M1

$= 0 \Rightarrow$ stationary value

CSO

shown = 0 (54 or 2×27 seen) and statement

A1

Substitute $t = 3$ into $\frac{d^2x}{dt^2} (= 14)$

M1

$\frac{d^2x}{dt^2} > 0 \Rightarrow$

minimum value

CSO; all values (if stated) must be correct

A1

4

(c) Substitute $t = 1$ into their $\frac{dx}{dt}$

must be their $\frac{dx}{dt}$ NOT $\frac{d^2x}{dt^2}$ etc

M1

$\frac{dx}{dt} = 28$

ft their $\frac{dx}{dt}$ when $t = 1$

A1ft

2

- (d) Substitute $t = 2$ into their $\frac{dx}{dt}$
 must be their $\frac{dx}{dt}$ NOT $\frac{d^2x}{dt^2}$ or x
 Interpreting their value of $\frac{dx}{dt}$

M1

$$= 16 - 80 + 66 = 2 \quad (> 0)$$

$\Rightarrow$ increasing when $t = 2$

Allow decreasing if their $\frac{dx}{dt} < 0$

E1ft

2

[13]

Q15.

- (a) (i) $p(2) = 8 + 2 - 10$
 Must find $p(2)$ NOT long division

M1

$\Rightarrow p(2) = 0 \Rightarrow (x - 2)$ is factor
 Shown = 0 plus a statement

A1

2

- (ii) Attempt at long division (generous)
 Obtaining a quotient $x^2 + cx + d$
 or equating coefficients (full method)

M1

$$p(x) = (x - 2)(x^2 + 2x + 5)$$

$a = 2, b = 5$ by inspection B1, B1

A1

2

(b) (i) $\frac{dy}{dx} = 3x^2 + 1$

One term correct

M1

All correct – no + c etc

A1

When $x = 2 \quad \frac{dy}{dx} = 3 \times 4 + 1$

Sub $x = 2$ into their $\frac{dy}{dx}$

m1

Therefore gradient at Q is 13
CSO

A1

4

(ii) $y = 13(x - 2)$

*Tangent (NOT normal) attempted
ft their gradient answer from (b)(i)*

M1

CSO; correct in any form

A1

2

(iii) $\int \dots dx = \frac{x^4}{4} + \frac{x^2}{2} - 10x (+c)$

one term correct

M1

second term correct

A1

all correct (condone no +c)

A1

3

(iv) $[4 + 2 - 20] - [0] = -14$

*F(2) attempted and possibly F(0)
Must have earned M1 in (b)(iii)*

M1

Area of shaded region = 14

CSO; separate statement following correct evaluation of limits

A1

2

[15]

Q16.

(a) (i) $\frac{dy}{dx} = 3x^{\frac{1}{2}}$
 $kx^{\frac{1}{2}}$ with or without + c

M1

= 6 {when $x = 4$ }

Must be 6 and seen in (a)(i)

6 + c is A0

A1cao

2

(ii) y-coordinate of A = $2 \times 4^{\frac{3}{2}} (= 16)$

Substitute $x = 4$ in $y = 2x^{\frac{3}{2}}$

M1

$6 \times m' = -1$

$m_1 \times m_2 = -1$ OE used with c's value of

$\frac{dy}{dx}$ when $x = 4$. PI

M1

$y - 16 = m(x - 4)$

dep on 1st M1 in (a)(ii)

m must be numerical

m1

$y - 16 = -\frac{1}{6}(x - 4)$
 ACF

A1

(b) (i) $\int 8x^{\frac{1}{2}} dx = \frac{8}{\frac{3}{2}} x^{\frac{1}{2}+1} \{+c\}$

Index raised by 1

M1

$$= \frac{16}{3} x^{\frac{3}{2}} \{+c\}$$

Condone missing '+ c'
Coefficient must be simplified

A1

2

(ii) $\int 2x^{\frac{3}{2}} dx = \frac{2}{\frac{5}{2}} x^{\frac{5}{2}} \{+c\} \quad \{= \frac{4}{5} x^{\frac{5}{2}} \{+c\}\}$

Can award for unsimplified form

B1

$$\int_0^4 8x^{\frac{1}{2}} dx - \int_0^4 2x^{\frac{3}{2}} dx$$

Ignore limits here

M1

$$= \frac{16}{3} (4)^{\frac{3}{2}} - 0 - \left[\frac{4}{5} (4)^{\frac{5}{2}} - 0 \right]$$

F(4) – F(0) used in either; {F(0) = 0 PI} rights reserved.
Cand. must be using F(x) as a result of
his/her integration in (b)(i) or in the (b)(ii)
B1 line above

M1

$$= \frac{256}{15}$$

Accept any value from 17.04 to 17.1
inclusive in place of 256/15

A1

4

(c) Translation

Accept 'translat...' as equivalent

[T or Tr is NOT sufficient]

B1

$$\begin{bmatrix} -3 \\ 0 \end{bmatrix}$$

*Accept equivalent in words provided
linked to 'translation/move/shift'
(B0B0 if >1 transformation)*

B1

2

[14]

Q17.

(a) $(1 + 2x)^4 = 1 + 4(2x) + 6(2x)^2 + 4(2x)^3 + (2x)^4$

M1

$$= 1 + 8x$$

$$+ 24x^2$$

$$+ 32x^2 \{+ 16x^4\}$$

*(1), 4, 6, 4, (1) OE unsimplified with
correct powers of x
Algebraic multiplication must be a full
method*

Accept a = 8 provided 1st term is 1

$$b = 24$$

$$c = 32$$

A1

A1

A1

4

(b) $(1 - 2x)^4 = 1 - 8x + 24x^2 - 32x^3 \{+ 16x^4\}$

Replace x by -x even in M1 line of (a) PI

M1

ft c's non zero values for a, b and c

A1Ft

$$(1 + 2x)^4 + (1 - 2x)^4$$

$$= 1 + 8x + 24x^2 + 32x^3 + 16x^4 + 1 - 8x + 24x^2 - 32x^3 + 16x^4$$

$$= 2 + 48x^2 + 32x^4$$

AG Be convinced

A1 cso

3

(c) $\frac{dy}{dx} = 96x + 128x^3$

A correct power of x OE

M1

For st. pt $96x + 128x^3 = 0$

A1

$$32x(3 + 4x^2) = 0$$

Since $3 + 4x^2 > 0$ there is only one stationary point

Any valid explanation of curve having just one stationary point

E1

The coordinates of the stationary point are (0, 2)

*(0, 2) as the **only** stationary point*

B1

4

[11]

Q18.

(a) $y = e^{2x}(x^2 - 4x - 2)$

$$\frac{dy}{dx} = e^{2x}(2x - 4)$$

Product rule; allow 1 slip

M1

$$+ (x^2 - 4x - 2)2e^{2x}$$

A1

$$\frac{dy}{dx} = e^{2x}(2x - 4 + 2x^2 - 8x - 4)$$

$$\text{Factorising } e^{2x}(2x^2 - 6x - 8)$$

M1

$$e^{2x}(2x^2 - 6x - 8)$$

$$\text{Or } x^2 - 3x - 4 = 0$$

A1

$$e^{2x} \neq 0$$

$$(x - 4)(x + 1) = 0$$

*Solving 3 term quadratic
Dependent on both M marks*

m1

$$x = 4, -1$$

And no extras eg $x = 0$

A1

6

(b) (i) $\frac{d^2 y}{dx^2} = e^{2x} \cdot 2 + (2x - 4)2e^{2x}$

Product rule from their $\frac{dy}{dx}$ in form
 e^{2x} (quadratic) $e^{2x}(4x^2 - 8x - 22)$

M1

$$+ (x^2 - 4x - 2)4e^{2x} + 2e^{2x}(2x - 4)$$

A1

OR

$$\frac{d^2 y}{dx^2} = e^{2x}(4x - 6) + (2x^2 - 6x - 8)2e^{2x}$$

M1A1

2

(ii) $x = 4 : y'' = e^8 (10) > 0 \therefore \text{MIN}$

Their 2 x 's in their $\frac{d^2 y}{dx^2}$
 only of form e^{2x} (quadratic)

M1

$$x = -1 : y'' = e^{-2} (-10) < 0 \therefore \text{MAX}$$

CSO Both correct

Allow values either side of y or y'

A1

2

[10]

Q19.

(a) $P(-1, \pi)$

Condone $(-1, 180^\circ)$

B1

$Q(1, 0)$

B1

2

(b) Translate

E1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

or equivalent in words

B1

Stretch SF 2 // y -axis

Stretch + one other correct

M1

all correct

A1

4

(c)



Correct shape in 1st quadrant

B1

2π and 2 marked correctly

B1

2

(d) (i) $\frac{y}{2} = \cos^{-1}(x - 1)$

M1

$$\cos\left(\frac{y}{2}\right) = x - 1$$

$$x = \cos\left(\frac{y}{2}\right) + 1$$

A1

(ii) $-\frac{1}{2} \sin\left(\frac{y}{2}\right)$
 $k \sin (...)$

M1

$\frac{dx}{dy}$ correct

A1

At $y = 2$, $\left(\frac{dx}{dy}\right) = -\frac{1}{2} \sin 1$

Condone AWRT -0.42

A1

3

[13]

Q20.

(a) $y = \frac{4x}{4x-3}$
 $\frac{dy}{dx} = \frac{(4x-3) \cdot 4 - 4x(4)}{(4x-3)^2}$

Must use quotient rule
 Condone one slip

M1

$= \frac{-12}{(4x-3)^2}$

$k = -12$

A1

2

(b) (i) $y = x \ln(4x-3)$
 $\frac{dy}{dx} = \frac{x \cdot 4}{4x-3} + \ln(4x-3)$

$\frac{f'(x)}{4x-3} + g(x)$ 'f(x)' may be constant

M1

$$\frac{kx}{4x-3} + \ln(4x-3)$$

m1A1

3

(ii) $x = 1 \quad y = 0$

B1

$$\frac{dy}{dx} = 4$$

Sub $x = 1$ into their $\frac{dy}{dx}$

M1

$\therefore y = 4(x - 1)$ any correct form
CSO Must have full marks in (b)(i)

A1

3

(c) (i) $u = 4x - 3$
 $du = 4 dx$

M1

$$\int \frac{4x}{4x-3} dx = \int \frac{u+3}{u} \frac{du}{4}$$

Or $\int \frac{4x}{4x-3} dx = \int \left(1 + \frac{3}{4x-3}\right) dx$

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A1

$$= \left(\frac{1}{4}\right) \int \left(1 + \frac{3}{u}\right) (du)$$

$$= \int \left(1 + \frac{3}{u}\right) du \text{ etc}$$

m1

$$= \frac{1}{4} (u + 3 \ln u)$$

$$= \frac{1}{4} [(4x-3) + 3 \ln(4x-3)] (+c)$$

CSO Condone missing du

A1

4

(ii) $\int \ln(4x - 3) \, dx$

$$u = \ln(4x - 3) \quad \frac{dv}{dx} = 1$$

In correct direction

M1

$$\frac{du}{dx} = \frac{4}{4x - 3} \quad v = x$$

$$\int x \ln(4x - 3) - \int \frac{4x}{4x - 3} \, dx$$

A1

$$= x \ln(4x - 3) - \frac{1}{4} [(4x - 3) + 3 \ln(4x - 3)]$$

x ln(4x - 3) - their (c)(i)

m1A1

(+ c)

4

[16]

Q21.

(a) $\frac{dy}{dx} = 5x^4 + 40x$

One of these powers correct

M1

One of these terms correct

A1

All correct (no + c etc)

A1

3

(b) $x = -2 \quad \frac{dy}{dx} = 5 \times (-2)^4 + (40 \times -2)$

Substitute $x = -2$ into their $\frac{dy}{dx}$

M1

$$\frac{dy}{dx} = 5 \times 16 + (40 \times -2) = 0 \Rightarrow P \text{ is stationary point}$$

*CSO Shown = 0 plus statement
 eg "st pt", "as required", "grad = 0" etc*

A1

Or *their* $\frac{dy}{dx} = 0 \Rightarrow x^2 = k$

(M1)

$$x^3 = -8 \Rightarrow x = -2$$

CSO $x = 0$ need not be considered

(A1)

2

(c) (i) $\frac{d^2y}{dx^2} = 20x^3 + 40$

Correct ft their $\frac{dy}{dx}$

B1

$$= 20 \times (-2)^3 + 40$$

Subst $x = -2$ into their second derivative

M1

$$(-160 + 40) = -120$$

CSO

A1

3

- (ii) Maximum (value)
 their c(i) answer must be < 0
 Other valid methods acceptable provided
 "maximum" is the conclusion
*Accept minimum if their c(i) answer > 0
 and correctly interpreted
 Parts (i) and (ii) may be combined by
 candidate but -120 must be seen to award
 A1 in part (c)(i)*

E1ft

1

(d) (When $x = 1$) $y = 13$

B1

When $x = 1$, $\frac{dy}{dx} = 5 + 40$

Sub $x = 1$ into their $\frac{dy}{dx}$

M1

$y = (\text{their } 45)x + k$ OE

ft their $\frac{dy}{dx}$

m1

Tangent has equation $y - 13 = 45(x - 1)$

CSO OE $y = 45x + c$, $c = -32$

A1

4

[13]

Q22.

(a) $\frac{dy}{dx} = \frac{45}{2}x^{\frac{1}{2}} - \frac{5}{2}x^{\frac{3}{2}}$

One power correctly obtained

M1

A1 for each term on the RHS coeffs simplified

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3

(b) $\frac{45}{2}x^{\frac{1}{2}} - \frac{5}{2}x^{\frac{3}{2}} = 0$

cand's (a) = 0

M1

$\frac{5}{2}x^{\frac{1}{2}}(9 - x) = 0$

Must be solving eqn of form $ax^m + bx^n = 0$,
 m and n non-zero, with at least one of m
 and n non-integer and reaching a stage
 from which the non-zero value of x can be

stated PI. Must deal with powers of x correctly and any squaring of kx^p terms or expressions must be correct.

m1

At M , $x = 9$

A1

$$y_M = 162$$

M1 must be scored, else 0/4

A1

4

(c) At $P(1, 14), \frac{dy}{dx} = \frac{45}{2} - \frac{5}{2} = 20$
Attempt to find $y'(1)$

M1

Tangent at P : $y - 14 = m(x - 1)$
 $m = \text{cand's value of } y'(1)$

m1

$$y - 14 = 20x - 20; y = 20x - 6$$

CSO; AG

A1

3

(d) Tangent at M : $y = 162$
ft $y = \text{cand's } y_M$

B1F

$$\text{At } R, 162 = 20x - 6; x = 8.4$$

Solving cand's numerical $y_M = 20x - 6$ to find a value for x

M1

$$\text{Distance } RM = |x_M - x_R| = 9 - 8.4 = 0.6$$

ft on coordinates of M

A1F

3

[13]

Q23.

(a) (i) $f(x) = \frac{\cos x}{2x+1} - \frac{1}{2}$

OE

$$f(0) = \frac{1}{2}; f\left(\frac{\pi}{2}\right) = -\frac{1}{2}$$

$$x=0 \quad LHS=1, \quad x=\frac{\pi}{2} \quad LHS=0$$

M1

Change of sign $0 < \alpha < \frac{\pi}{2}$

Either side of $\frac{1}{2}, \therefore 0 < \alpha < \frac{\pi}{2}$

A1

2

(ii) $\frac{\cos x}{2x+1} = \frac{1}{2}$

$$\left. \begin{array}{l} 2\cos x = 2x+1 \\ 2\cos x - 1 = 2x \end{array} \right\} \text{or, } \cos x = x + \frac{1}{2}$$

Either line

$$x = \cos x - \frac{1}{2}$$

$$\cos x - \frac{1}{2} = x$$

AG; or

All correct with no errors

B1

1

(iii) $x_1 = 0$
 $x_2 = 0.5$

Attempt at iteration

(allow $x_2 = -0.5, x_3 = 0.38, 0.4$)

M1

$$x_3 = 0.378$$

CAO

A1

2

(b) (i) $\frac{dy}{dx} = \frac{(2x+1)(-\sin x) - \cos x \times 2}{(2x+1)^2}$

Attempt at quotient rule:

$$\frac{\pm (2x+1)\sin x \pm 2\cos x}{(2x+1)^2}$$

M1

Either term correct

A1

All correct ISW

A1

3

(ii) $x = 0$

$$\frac{dy}{dx} = -2$$

Correctly subst. $x = 0$ into their $\frac{dy}{dx}$

m1

$\therefore$ Gradient of normal = $\frac{1}{2}$
CSO

A1

2

EXAM PAPERS PRACTICE

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[10]

Q24.

(a) $\int (t-1)\ln t \, dt$

$$u = \ln t \quad \frac{dv}{dt} = t - 1$$

Differentiate + integrate, correct direction

M1

$$\frac{dz}{dt} = \frac{1}{t} \quad v = \frac{t^2}{2} - t$$

All correct

A1

$$\begin{aligned}
 \int \left(\frac{t^2}{2} - t \right) \ln t - \left(\frac{t^2}{2} - t \right) \times \frac{1}{t} (dt) \\
 = \left(\frac{t^2}{2} - t \right) \ln t - \int \left(\frac{t}{2} - 1 \right) (dt)
 \end{aligned}$$

Condone missing brackets

A1

$$= \left(\frac{t^2}{2} - t \right) \ln t - \frac{t^2}{4} + t (+c)$$

CAO

A1

4

Alternative

$$\int (t-1) \ln t$$

$$u = \ln t \quad v' = (t-1)$$

(M1)

$$u' = \frac{1}{t} \quad v = \frac{(t-1)^2}{2}$$

(A1)

$$\int = \frac{(t-1)^2}{2} \ln t - \int \frac{(t-1)^2}{t} \frac{1}{t} dt$$

$$\frac{(t-1)^2}{2} \ln t - \frac{1}{2} \int \frac{t^2 - 2t + 1}{t} dt$$

$$\frac{(t-1)^2}{2} \ln t - \frac{1}{2} \int t - 2 + \frac{1}{t} dt$$

(A1)

$$\frac{(t-1)^2}{2} \ln t - \frac{1}{2} \left[\frac{t^2}{2} - 2t + \ln t \right]$$

(A1)

$$= \frac{t^2}{2} \ln t - t \ln t + \frac{1}{2} \ln t - \frac{t^2}{4} + t - \frac{1}{2} \ln t$$

$$= \left(\frac{t^2}{2} - t \right) \ln t - \frac{1}{4} t^2 + t + c$$

(4)

(b) $t = 2x + 1$
 $dt = 2 dx$ (RHS)

$$\frac{dt}{dx} = 2 \quad (LHS)$$

M1

$2x = t - 1,$
 OE

m1

$$\int = \int 2(t-1) \ln t \frac{dt}{2}$$

AG

A1

3

(c) $[x]_1^3 = [t]_1^3$

Limit becoming 3

M1

$$\int = \left[\left(\frac{t^2}{2} - t \right) \ln t - \frac{t^2}{4} + t \right]_1^3$$

$$= \left[\left(\frac{9}{2} - 3 \right) \ln 3 - \frac{9}{4} + 3 \right] - \left[0 - \frac{1}{4} + 1 \right]$$

Correctly sub. 1, 3 into their (a)

m1

$$= \frac{3}{2} \ln 3$$

CSO

A1

3

or

$$\int = \left[\left(\frac{(2x+1)^2}{2} - (2x+1) \right) \ln (2x+1) - \frac{(2x+1)^2}{4} + (2x+1) \right]_0^1$$

Condone 1 slip

(M1)

$$= \left(\left(\frac{9}{2} - 3 \right) \ln 3 - \frac{9}{4} + 3 \right) - \left(0 - \frac{1}{4} + 1 \right)$$

Correctly sub. 0, 1

(m1)

$$= \frac{3}{2} \ln 3$$

CSO

(A1)

(3)

[10]

Q25.

(a) $\frac{dy}{dx} = 4x^3 - 32$

Reduce one power by 1

M1

One term correct

A1

All correct (no + c etc)

A1

3

(b) Stationary point $\Rightarrow \frac{dy}{dx} = 0$

M1

$$\Rightarrow x^3 = 8$$

$$x^n = k \text{ following from their } \frac{dy}{dx}$$

A1ft

$$\Rightarrow x = 2$$

CSO

A1

3

(c) (i) $\frac{d^2y}{dx^2} = 12x^2$

FT their $\frac{dy}{dx}$

B1ft

1

(ii) When $x = 2$, $\frac{d^2y}{dx^2}$ considered

Or complete test with $2 \pm \varepsilon$ using $\frac{dy}{dx}$

M1

$\Rightarrow$ minimum point

E1ft

2

(d) Putting $x = 0$ into their $\frac{dy}{dx}$ ($= -32$)

M1

$\frac{dy}{dx} < 0 \Rightarrow$ decreasing

Allow "increasing" if their $\frac{dy}{dx} > 0$

A1ft

2

EXAM PAPERS PRACTICE

[11]

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Q26.

(a) (i) $p(-1) = -1 + 7 - 6$
Finding $p(-1)$

M1

$= 0$ therefore $x + 1$ is a factor
Shown to $= 0$ **plus statement**

A1

2

(ii) $p(x) = (x + 1)(x^2 - x - 6)$
Long division/inspection (2 terms correct)

M1

Quadratic factor correct

A1

$$p(x) = (x + 1)(x + 2)(x - 3)$$

May earn M1, A1 For correct second factor
then A1 For $(x + 1)(x + 2)(x - 3)$

A1

3

(b) (i) $A(-2, 0)$

Condone $x = -2$

B1

1

(ii) $\frac{x^4}{4} - \frac{7x^2}{2} - 6x (+c)$

One term correct

M1

Another term correct

A1

(may have + c or not)

All correct unsimplified

A1

$$\left[\frac{81}{4} - \frac{63}{2} - 18 \right] - \left[\frac{1}{4} - \frac{7}{2} + 6 \right]$$

$F(3) - F(-1)$ attempted in correct order

m1

$$= -32$$

CSO; OE

A1

5

(iii) Area of shaded region = 32

FT their (b)(ii) but positive value needed

B1ft

1

(iv) $\frac{dy}{dx} = 3x^2 - 7$

One term correct

M1

All correct (no + c etc)

A1

When $x = -1$, gradient = -4

CSO

A1

3

(v) Gradient of normal = $\frac{1}{4}$

B1ft

$y = \text{"their gradient"} (x \pm 1)$

*Must be finding **normal**, not tangent*

M1

$$y = \frac{1}{4}(x + 1)$$

CSO; any correct form eg $4y - x = 1$

A1

3

[18]

Q27.

(a) (i) $\frac{dy}{dx} = 4 \times \frac{1}{2} x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}} = 2x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}}$

A power decreased by 1

M1

A1 For each correct term

A1 A1

3

(ii) At $P(4, 0)$, $\frac{dy}{dx} = \frac{2}{\sqrt{4}} - \frac{3}{2} \times 2$

Attempts $\frac{dy}{dx}$ when $x = 4$

M1

$$= 1 - 3 = -2$$

AG

A1

2

(iii) Gradient of normal = $\frac{1}{2}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 0 = m [x - 4]$

m numerical; can be awarded even if $m = -2$

M1

$$y - 0 = \frac{1}{2}(x - 4) \Rightarrow 2y = x - 4$$

ACF of the equation

A1

3

(iv) At Q, $x = 0$, $2y = 0 - 4$
PI

M1

$$y_e = -2$$

Area of triangle $OPQ = 0.5 \times 4 \times |y_e|$

Ft on a linear equation for normal provided y_e is negative and prev A1 is lost

A1F

$$= 4$$

Ft on c's negative y_e

B1F

3

(v) $2x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} = 0 \Rightarrow 2x^{-\frac{1}{2}} = \frac{3}{2}x^{\frac{1}{2}}$

Puts c's $\frac{dy}{dx} = 0$ and a 1st step in attempt to solve.

M1

$$2 = \frac{3}{2}x \quad \Rightarrow x = \frac{4}{3}$$

Valid method to $ax = b$

m1

Condone 1.3 or better

A1

3

(b) (i)

$$\int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx = 4 \frac{x^{\frac{3}{2}}}{1.5} - \frac{x^{\frac{5}{2}}}{2.5} \{+c\}$$

$$= \frac{8}{3} x^{\frac{3}{2}} - \frac{2}{5} x^{\frac{5}{2}} \{+c\}$$

One power correct

M1

Condone absence of "+ c"

A1,A1

3

(ii) Area under curve =

$$4 \frac{x^{\frac{3}{2}}}{1.5} - \frac{x^{\frac{5}{2}}}{2.5} - \{0\}$$

$F(4) - \{F(0)\}$

M1

Total area = $F(4)$ + area triangle OPQ

m1

$$\text{Total area} = \frac{128}{15} + 4 = \frac{188}{15} = 12.5(3...)$$

Accept 3 sf if clear

A1

3

[20]

Q28.

(a) (i) $y = (2x^2 - 5x + 1)^{20}$

$$\frac{dy}{dx} = 20(2x^2 - 5x + 1)^{19} (4x - 5) \quad \text{OE}$$

chain rule 20 ()¹⁹ f(x)

M1

with no further incorrect working

A1

2

(ii) $y = x \cos x$

$$\frac{dy}{dx} = -x \sin x + \cos x$$

product rule $\pm x \sin x \pm \cos x$

M1

CSO

A1

2

(b) $y = \frac{x^3}{x-2}$

$$\frac{dy}{dx} = \frac{(x-2)3x^2 - x^3 \times 1}{(x-2)^2}$$

quotient rule $\frac{\pm uv' \pm u'v}{(x-2)^2}$

condone missing brackets

M1

A1

$$= \frac{3x^2 - 6x^2 - x^3}{(x-2)^2}$$

$$= \frac{2x^2(x-3)}{(x-2)^2}$$

CSO

A1

3

[7]

Q29.

(a) $V = x(9 - 3x)^2$

*Attempt at V in terms of x (condone slip
when rearranging formula for $y = 9 - 3x$)
or $(9 - 3x)^2 = 81 - 54x + 9x^2$*

M1

$$V = x(81 - 54x + 9x^2)$$

$$= 81x - 54x^2 + 9x^3$$

AG; no errors in algebra

A1

2

(b) (i) $\frac{dV}{dx} = 81 - 108x + 27x^2$

One term correct

M1

Another correct

A1

All correct (no + c etc)

A1

$$= 27(x^2 - 4x + 3)$$

CSO; all algebra and differentiation correct

A1

4

(ii) $(x - 3)(x - 1)$ or $(27x - 81)(x - 1)$ etc

"Correct" factors or correct use of formula

M1

$$\Rightarrow x = 1, 3$$

SC: B1, B1 For $x = 1$, $x = 3$ found by inspection (provided no other values)

A1

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2

(c) $\frac{d^2V}{dx^2} = -108 + 54x$ (condone one slip)

ft their $\frac{dV}{dx}$ (may have cancelled 27 etc)

M1

CSO; all differentiation correct

A1

2

(d) (i) $x = 3 \Rightarrow \frac{d^2V}{dx^2} = 54; \quad x = 1 \Rightarrow \frac{d^2V}{dx^2} = -54$

ft their $\frac{d^2V}{dx^2}$ and their **two** x -values

B1ft

1

(ii) $(x =) 1$ (gives maximum value)

Provided their $\frac{d^2V}{dx^2} < 0$

E1

1

(iii) $V_{\max} = 36$
CAO

B1

1

[13]

Q30.

(a) $\sqrt{x^3} = x^{\frac{3}{2}}$

OE; accept 'k = 1.5'

B1

1

(b) (i) $\frac{dy}{dx} = 2x - \frac{3}{2}x^{\frac{1}{2}}$

At least one index reduced by 1 and no term of the form $\sqrt{ax^2}$.

M1

For $2x$

B1

For $-1.5x^{0.5}$. Ft on ans (a) non-integer k

A1F

3

(ii) When $x = 4$, $y = 8$

B1

$y'(4) = ;$

$\frac{dy}{dx}$
 Attempt to find $\frac{dy}{dx}$ when $x = 4$

M1

$$= 2(4) - 1.5(\sqrt{4}) = 5$$

Ft on one earlier error provided noninteger powers in (a) and (b) (i)

A1F

Tangent: $y - 8 = 5(x - 4)$

$$y - y(4) = y'(4)[x - 4] \text{ OE}$$

m1

$$y = 5x - 12$$

CSO; must be $y = 5x - 12$

A1

5

[9]

Q31.

(a) $\frac{dy}{dx} = \frac{5(3x+1)^4 \times 3}{k(3x+1)^4}$

M1

$$= 15(3x+1)^4$$

with no further errors (w.n.f.e)

A1

2

(b) $\frac{dy}{dx} = \frac{3}{3x+1}$

$$\frac{k}{3x+1}$$

M1

w.n.f.e

A1

2

(c) $\frac{dy}{dx} = (3x+1)^5 \times \frac{3}{3x+1} + \ln(3x+1) \times 15(3x+1)^4$

product rule $uv' + u'v$ (from (a) and (b))

M1

either term correct

A1

CSO with no further errors

A1

$$\left(\begin{aligned} &= (3x+1)^4 [3 + 15 \ln(3x+1)] \\ &= 3(3x+1)^4 [1 + 5 \ln(3x+1)] \end{aligned} \right)$$

3

[7]

Q32.

(a) $\frac{dy}{dx} = -x^2 \sin 2x + \cos 2x$

product rule $kx \sin 2x \pm \cos 2x$

M1

no further incorrect working

A1

2

(b) (i) $-2\alpha \sin 2\alpha + \cos 2\alpha = 0$

replacing $x = \alpha$ and writing equation
equal to zero (at any line)

M1

$$\left. \begin{aligned} 2\alpha \sin 2\alpha &= \cos 2\alpha \\ 2\alpha \tan 2\alpha &= 1 \end{aligned} \right\} \text{either}$$

$$2\alpha \tan 2\alpha - 1 = 0$$

AG; CSO

A1

2

(ii) $\left. \begin{aligned} f(0.4) &= 0.2 \\ f(0.5) &= -0.6 \end{aligned} \right\} \text{awrt} \quad \text{o.e.}$

(0.9's unsubstantiated scores M0)

M1

Change of sign $\therefore 0.4 < \alpha < 0.5$

A1

2

(iii) $2x \tan 2x = 1$

$$\left. \begin{array}{l} \tan 2x = \frac{1}{2x} \\ 2x = \tan^{-1}\left(\frac{1}{2x}\right) \end{array} \right\} \text{either}$$

$$x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$$

AG; CSO

B1

1

(iv) $x_1 = 0.4$
 $x_2 = 0.4480\dots$
 $x_2 = 25.7$

M1

$x_3 = 0.4200\dots$
 $= 0.42$

A1

2

(c) $y = x \cos 2x$

$$\left. \begin{array}{l} u = x \quad du = 1 \\ dv = \cos 2x \quad v = \frac{\sin 2x}{2} \end{array} \right\}$$

$\left. \begin{array}{l} \text{differentiate one term} \\ \text{integrate one term} \end{array} \right\} \text{ must be } k \sin 2x$

M1

*correct substitution of their values into
 parts formula using $u = x$*

m1

$$\int = \frac{x \sin 2x}{2} - \int \frac{\sin 2x}{4} (dx)$$

$$= \left[\frac{x \sin 2x}{2} + \frac{\cos 2x}{4} \right]_{(0)}^{(0.5)}$$

A1

$$= \left(\frac{\sin 1}{4} + \frac{\cos 1}{4} \right) - \left(\frac{\cos 0}{4} \right)$$

*correctly substituting values from
previous 2 method marks*

m1

$$= 0.0954$$

AWRT

A1

5

[14]

Q33.

(a) (i) $2x^2 + 2xh + 4xh \quad (= 54)$

Attempt at surface area (one slip)

M1

$$\Rightarrow x^2 + 3xh = 27$$

AG CSO

A1

2

(ii) $h = \frac{27 - x^2}{3x} \text{ or } h = \frac{9}{x} - \frac{x}{3} \text{ etc}$

Any correct form

B1

1

(iii) $V = 2x^2h = 18x - \frac{2x^3}{3}$

AG (watch fudging) condone omission of brackets

B1

1

(b) (i) $\frac{dV}{dx} = 18 - 2x^2$

One term correct "their" V

M1

All correct unsimplified $18 - 6x^2/3$

A1

2

(ii) Sub $x = 3$ into their $\frac{dV}{dx}$

Or attempt to solve their $\frac{dV}{dx} = 0$

M1

Shown to equal 0 plus **statement** that this implies a stationary point if verifying

CSO Condone $x = \pm 3$ or $x = 3$ if solving

A1

2

(c) $\frac{d^2V}{dx^2} = -4x$

(= -12)

FT their $\frac{dV}{dx}$

B1 ✓

$\frac{d^2V}{dx^2} < 0$

at stationary point $\Rightarrow$ maximum

FT their second derivative conclusion

If "their" $\frac{d^2y}{dx^2} > 0 \Rightarrow$ minimum etc

E1 ✓

2

[10]

Q34.

(a) (i) $B(0, 5)$

B1

Area AOB $\frac{1}{2} \times 1 \times 5$

Condone slip in number or a minus sign

M1

$= 2\frac{1}{2}$

A1

(ii) $\frac{3x^6}{6} + \frac{2x^2}{2} + 5x$ or $\frac{x^6}{2} + x^2 + 5x$ (may have + c or not)

Raise one power by 1

M1

One term correct

A1

All correct unsimplified

A1

(iii) Area under curve = $\int_{-1}^0 f(x) dx$
Correctly written or $F(0) - F(-1)$ correct

B1

$[0] - \left[\frac{1}{2} + 1 - 5 \right]$

*Attempt to sub limit(s) of -1 (and 0)
 Must have integrated*

M1

Area under curve $3\frac{1}{2}$
CSO (no fudging)

A1

Area of shaded region = $3\frac{1}{2} - 2\frac{1}{2} = 1$

FT their integral and triangle (very generous)

B1 ✓

(b) (i) $\frac{dy}{dx} = 15x^4 + 2$

One term correct

M1

All correct (no +c etc)

A1

when $x = -1$, gradient = 17

CSO

A1

3

(ii) $y = \text{"their gradient"} (x + 1)$

*Must be finding **tangent** – not normal
any form e.g. $y = 17x + 17$*

B1 ✓

1

[14]

Q35.

(a) (i) $y = x + 1 + 4x^{-2} \Rightarrow \frac{dy}{dx} = 1 - 8x^{-3}$

M1

Power $p \rightarrow p - 1$

(A1 if $1 + ax^n$ with $a = -8$ or $n = -3$)

A2,1,0

3

(ii) $1 - 8x^{-3} = 0$

Puts c's $\frac{dy}{dx} = 0$

M1

$x^3 = 8$

Using $x^{-k} = \frac{1}{x^k}$ to reach $x^a = b$, $a > 0$ or correct use of logs.

m1

$x = 2$

A1

When $x = 2$, $y = 4$

A1ft

4

(iii) At (1, 6), $\frac{dy}{dx} = 1 - 8 = -7$

Attempt to find $y'(1)$

M1

Gradient of normal = $\frac{1}{7}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 6 = m[x - 1]$

m numerical

M1

$$y - 6 = \frac{1}{7}(x - 1)$$

$$\left\{ \frac{y - 6}{x - 1} = \frac{1}{7}; 7y = x + 41 \right\}$$

OE ft on c's answer for (a)(i) provided at least A1 given in (a)(i) and previous 3M marks awarded

Alt

4

(b) (i) $\int x \left(1 + \frac{4}{x^2} \right) dx =$

$$\dots\dots\dots = \frac{x^2}{2} + x - 4x^{-1} \{+c\}$$

One of three terms correct.

M1

One of three terms correct.

For A2 need all three terms as printed or better

(A1 if 2 of 3 terms correct)

A2,1,0

3

(ii) {Area = } $\int_1^4 x + 1 + \frac{4}{x^2} dx =$

$$\left[\frac{x^2}{2} + x - \frac{4}{x} \right]_1^4 = (8 + 4 - 1) - \left(\frac{1}{2} + 1 - 4 \right)$$

Dealing correctly with limits; $F(4) - F(1)$
(must have integrated)

= 13.5

M1

A1

2

[16]

